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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGY201 - VALUE ADDED PRODUCTS OF ANIMALS

(2024 ADMISSION ONWARDS)

ANSWER KEY

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 marks

1. Controlled crystallization in creamed honey is achieved by adding _____. [K] / [CO1]

   b) Seed honey

2. Chevon refers to the meat of: [K] / [CO3]

   a) Goat **(Out of syllabus. Give full credit, if attended)**

3. Which type of packaging is most common in canned meat? [U] / [CO3]

   b) Tinplate cans

4. Which bacteria are mainly responsible for lactic acid production : [U] / [CO4]

   b) Lactobacillus

5. What role does nitrogen (N_2) play in MAP? [U] / [CO3]

c) Inhibits microbial growth

6. The dried fruits used in preparing honey with fruits and nuts are ..... [K] / [CO1]

d) All of the above

7. Which chitosan property helps in cholesterol control? [U] / [CO2]

a) Fat binding ability

8. Drying of meat is one of the [K] / [CO3]

a) Oldest preservation methods.

9. Which milk product is commonly used to make Rasgulla Mix? [U] / [CO4]

a) Paneer

10. The type of pasteurization suitable for continuous processing of large volumes of milk] / [CO4]

b) HTST

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries 3 marks

11. Why does honey not spoil easily? [U] / [CO1]

Honey has an antibacterial effect because of 1) **high acidity** (pH 3 to 4.5), 2) **low water activity** (hypertonic), 3) **antibacterial substances** like phenolic acid, flavonoids, hydrogen peroxide etc., 4) **antibacterial enzymes** (glucose oxidase). (**Any 3 points, give full marks)**

12. Explain why gelatin is widely used as a stabilizing and emulsifying agent in foods. [U] / [CO2]

Gelatin is widely used as a stabilizing and emulsifying agent in foods due to its

amphoteric properties (containing both hydrophilic and hydrophobic regions), its **ability to form a thermo-reversible gel network**, its water binding capacity, ability to inhibit the formation of ice crystals, and its ability to help **foam formation**. **(Liberal Valuation)**

13. Explain the importance of liver as a food and pharmaceutical product. [An] / [CO3]

As Food Product:- It is one of the most nutrient-dense foods, packed with essential vitamins (vitamin A and B complex- particularly B12), minerals (iron,copper, zinc, phosphorus, and selenium etc) , and high-quality protein (containing all essential amino acids). **As Pharmaceutical Product**:- as a dietary supplement, for isolating anticoagulant heparin, its cellular components and enzymes are utilized in drug metabolism research and development. (**Any two points from each category. 1.5 Marks each; Liberal Valuation**)

14. Briefly explain the preparation of traditional Indian sweets like peda and burfi. [U] / [CO4]

Peda:- Broken the Koha into bits, Mix with sugar, Put into a Pan and cook over a very slow fire, pour into a greased tray. After cool and set, cut into desired size and shape. **Burfi**:- 4 to 5 litres of buffalo milk is heated in a stainless-steel kettle with constant stirring to prevent scorching. When semisolid consistency is reached, stop heating, and add powdered sugar, mix well using a wooden ladle until uniform, smooth mass is formed. Then, pour the mixture into a greased tray. After cool and set, cut into desired size and shape.

(Preparation of Peda-1.5 marks and Preparation of Burfi- 1.5 marks).

15. Define prepared meat products with two examples [K] / [CO3]

Prepared meat or **processed meat** products are any meats that have been modified from their fresh state to either improve their flavor or extend their shelf life (through multiple processing steps). Examples: Meatballs, Sausages, Bacon (made from pork belly or the back of the pig), deli meats, pre-cooked roasts, meat pies, ready-meals containing meat.

(**Definition- 1 Mark, Any two examples- 2 Marks (1 mark each**)

16. State the main property of honey useful in wound healing. [K] / [CO1]

The main property of honey useful in wound healing is its **antimicrobial activity**, which is due to its **high sugar content**, **low pH**, and **hydrogen peroxide production**. **(any 3 points- give full marks)**

17. Discuss the advantages of FPC as a protein supplement in human diets with suitable examples. [U] / [CO2]

Fish Protein Concentrate (FPC) has a high content of proteins, essential amino acids (lysine), Omega-3 Fatty Acids, and essential minerals, **(any 3 advantages; 1 mark each).**

18. Compare the storage requirements of liquid, frozen, and powdered eggs. [An] / [CO3]

Liquid Eggs:- Packaged in cartons, plastic pouches, or bulk containers, then refrigerated at 0 to 4^0C. **Frozen Eggs**:- Stored in containers (eg. Pails, drums) in freezers at -18^0C or lower. **Powdered Eggs**:- Packaged in moisture -proof, airtight containers (eg. Bags, drums, nitrogen-flushed pouches) to prevent oxidation and moisture absorption. Stored in cool, dry condition at below 20^0C. **(1 Mark for each liquid, frozen, and powdered eggs)**

19. Define 'Chakka' and mention its role in Srikhand preparation. [K] / [CO4]

Chakka (strained yogurt) is a semi-solid, concentrated milk product obtained by partially or completely draining the whey from plain dahi or yogurt. It has a smooth, uniform, semi-solid consistency and is typically white to pale yellow in color. Chakka is the primary and base ingredient for the preparation of the Shrikhand, which is a semi-soft, concentrated, sweetish-sour form of fermented milk. **(Definition- 1.5 Marks, mentioning role in Srikhand- 1.5 Marks)**

20. How is wort fermented for preparing Honey beer? [U] / [CO1]

Honey beer preparation begins with the preparation of wort by boiling barley malt and hops to extract fermentable sugars and flavor. To ferment wort for honey beer, the hot wort is first cooled and then honey is added. Then, yeast is added for fermentation (7 to 14 days).

21. What is beeswax? [K] / [CO1]

Beeswax is a natural wax produced by honeybees by using their four pairs of wax glands situated on the anterior part of the worker's last four normal sternal plates. Bees secrete it as a liquid that hardens into scales upon contact with air, and they use these scales to build the hexagonal cells of their honeycomb. It is composed mainly of esters of fatty acids and long-chain alcohols, free fatty acids, hydrocarbons and other minor components (eg. Carotenoids, propolis etc.). **(Definition-2 Marks, Composition- 1 mark)**

22. State why deoiling is required in the production of fish silage from oily fishes. [K] / [CO2]

Deoiling is required in the production of fish silage from oily fish primarily to **prevent the oxidation** or rancidity, Maintains **Nutritional Value**, Improves **Product Stability** and **Palatability**.

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries 5 marks

23. Give a short essay on secondary processing in Poultry meat production. [K] / [CO3]

    Secondary processing in poultry meat production refers to the transformation of carcasses or basic cuts into value-added products through various techniques like 1) cut-up & deboning, 2) portioning & trimming, 3) grinding & comminution, 4) forming (shaping), 5) Coating and cooking, and 6) chilling/freezing. (**0.5 mark for each techniques**)

24. Discuss the applications and advantages of honey caramels. [U] / [CO1]

    Honey caramels applications:-in Confectionery, as Toppings and Sauces, as Flavoring Agent, as Medicinal/Health Foods. **Advantages**: contains Natural Sweetener, Rich Antioxidants, trace amounts of Vitamins, Minerals and Amino Acids, Antimicrobial agents etc. (**Liberal valuation, 2.5 Mark for applications and 2.5 Marks for advantages**)

25. Explain the methods of extraction and preservation of fish glue. [U] / [CO2]

    Fish glue is a protein-based adhesive made from collagen extracted from fish by-products like skin, bones, and swim bladders. **Methods of Extraction**:-Raw Material Preparation, Pre-treatment (Soaking and Cleaning), Heating/Decoction, Filtration, Concentration and Drying. **Methods of Preservation**:- Drying, Chemical Preservation, Low-Temperature Storage. **(Extraction- 2.5 Marks, Preservation- 2.5 Marks)**

26. Explain the process of honey production in honeybees. [K] / [CO1]

    Honeybees collect nectar from flowers, storing in stomach where enzymes begin to convert it into honey, deposit the partially-transformed nectar into honeycomb cells, fan it with their wings to evaporate excess water until it becomes thick honey, which is then sealed with a wax cap. **(1 Mark to each step)**

27. What are the main environmental challenges of using fish waste fertilizer, and how [U] / [CO2]

can they be solved?

Improper disposal leading to pollution from odors, pathogens, and nutrient overload in water bodies (eutrophication). These can be solved by processing the waste into fertilizer through composting or other methods like anaerobic digestion, which reduces pollution and creates a valuable resource. **(Liberal valuation, challenges- 2.5 Marks, solution- 2.5 Marks)**

28. Explain the nutritional significance of eggs. Describe different egg products and its advantages [U] / [CO3]

Nutritional significance: - Source of high-quality proteins, essential amino acids, vitamins and minerals. **Different egg products**: - Liquid eggs, frozen eggs, dried egg products (whole egg powder, egg yolk powder, egg albumin powder). **Advantages**: - Convenience, extended shelf life, improved food safety, easier to incorporate into recipes. **(nutritional significance- 2 Marks, Different egg products- 2 Marks, advantages- 1 Mark)**

29. Describe the uses of the product 'Honey with fruits and nuts'. [U] / [CO1]

Honey with fruits and nuts is used as a food item for snacking, breakfast, and cooking, and for its various health benefits **(Liberal valuation)**

30. Describe the preservation techniques used in RTC & RTE products [U] / [CO2]

Preservation Techniques for Ready-to-Cook (RTC) Products:- Refrigeration/Chilling, Washing and Sanitation, Adding antimicrobial agents, Modified Atmosphere Packaging (MAP), Irradiation, Freezing or Dehydration, or Drying. **Preservation Techniques for Ready-to-Eat (RTE) Products**: - thermal sterilization, Modified Atmosphere Packaging (MAP), High-Pressure Processing (HPP), Rapid Freezing, adding approved chemical preservatives, Vacuum Cooling or Packaging. **(Liberal Valuation, RTC- 2.5 Marks, RTE- 2.5 Marks)**

[5x6 = 30]